



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2020 Year 12 Course Assessment Task 4 (Trial Examination)

Thursday August 20, 2020

General instructions

- Working time – 3 hours.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 13)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MXX.1 – Ms Ham

☐ 12MXX.2 – Mr Lam

Marker's use only.

QUESTION	1-10	11	12	13	14	15	16	%
MARKS	$\overline{10}$	$\overline{13}$	$\overline{14}$	$\overline{14}$	$\overline{18}$	$\overline{14}$	$\overline{17}$	$\overline{100}$

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided (labelled as page 13).

Questions

Marks

1. Consider the following statement for $n \in \mathbb{Z}$: 1

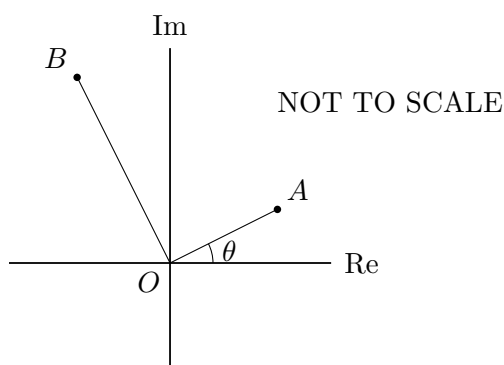
If $n^2 + 4n + 1$ is even, then n is odd

Which of the following statements is the contrapositive of this statement for $n \in \mathbb{Z}$?

- (A) If n is even, then $n^2 + 4n + 1$ is odd. (C) If n is odd, then $n^2 + 4n + 1$ is even.
 (B) If $n^2 + 4n + 1$ is odd, then n is even. (D) If $n^2 + 4n + 1$ is even, then n is even.
2. A particle is moving in simple harmonic motion about a fixed point O on a line. 1
 At time t seconds, it has displacement $x = 2 \cos(\pi t)$ metres from O .

What is the time taken by the particle to travel the first 100 metres of its motion?

- (A) 20 seconds (C) 50 seconds
 (B) 25 seconds (D) 100 seconds
3. The points A and B in the diagram represent the complex numbers z_1 and z_2 respectively, where $|z_1| = 1$ and $\text{Arg}(z_1) = \theta$ and $z_2 = \sqrt{3}iz_1$. 1



Which of the following represents $z_2 - z_1$?

- (A) $2e^{i(\frac{2\pi}{3} + \theta)}$ (B) $3e^{i(\frac{2\pi}{3} + \theta)}$ (C) $2e^{i(\frac{2\pi}{3} - \theta)}$ (D) $3e^{i(\frac{2\pi}{3} - \theta)}$

4. A particle of mass m kilograms has acceleration $a \text{ ms}^{-2}$ proportional to the square of its velocity $v \text{ ms}^{-1}$, i.e. 1

$$ma = -kv^2$$

for some positive constant k .

Which of the following integrals will result in a relationship between the time in seconds and velocity v ?

(A) $t = \int \frac{-m}{kv^2} dv$ (B) $t = \int \frac{m}{kv^2} dv$ (C) $t = \int \frac{-k}{mv^2} dv$ (D) $t = \int \frac{k}{mv^2} dv$

5. If ω is a complex cube root of unity of *smallest* positive argument, which of the following is the value of $\left(1 + \frac{1}{\omega}\right)^{2020}$? 1

(A) ω (B) $-\omega$ (C) 0 (D) 1

6. The points A , B and C are collinear where $\overrightarrow{OA} = \underline{i} + \underline{j}$, $\overrightarrow{OB} = 2\underline{i} - \underline{j} + \underline{k}$ and $\overrightarrow{OC} = 3\underline{i} + a\underline{j} + b\underline{k}$. 1

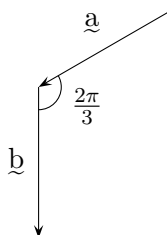
What are the values of a and b ?

(A) $a = -3, b = -2$ (C) $a = -3, b = 2$
(B) $a = 3, b = -2$ (D) $a = 3, b = 2$

7. Which of the following is an expression for $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x} dx$ after using the substitution $t = \tan \frac{x}{2}$? 1

(A) $\int_0^1 \frac{1}{1 + 2t} dt$ (B) $\int_0^1 \frac{2}{1 + 2t} dt$ (C) $\int_0^1 \frac{1}{(1 + t)^2} dt$ (D) $\int_0^1 \frac{2}{(1 + t)^2} dt$

8. In the following diagram, the vectors $\underline{a}$ and $\underline{b}$ are related such that $|\underline{a}| = |\underline{b}|$. 1



Given $|\underline{a}| = a$, which of the following expressions is equal to $\underline{a} \cdot \underline{b}$?

(A) $-\frac{\sqrt{3}}{2}a^2$ (B) $-\frac{1}{2}a^2$ (C) $\frac{1}{2}a^2$ (D) $\frac{\sqrt{3}}{2}a^2$

9. Which of the following complex numbers is a 6th root of i ? 1

(A) $-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$ (B) $-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$ (C) $-\sqrt{2} + \sqrt{2}i$ (D) $-\sqrt{2} - \sqrt{2}i$

10. Which integral is necessarily equal to $\int_{-a}^a f(x) \, dx$? 1

(A) $\int_0^a (f(x) - f(-x)) \, dx$

(C) $\int_0^a (f(x - a) - f(-x)) \, dx$

(B) $\int_0^a (f(x) - f(a - x)) \, dx$

(D) $\int_0^a (f(x - a) + f(a - x)) \, dx$

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (13 Marks)	Commence a NEW booklet.	Marks
(a) A three digit numeral n has x , y and z as its digits when writing from left to right respectively.		2
For example, if $n = 314$, then $x = 3$, $y = 1$ and $z = 4$.		
Show that if $x + z = y$, then the number is divisible by 11.		
(b) If $f^{(n)}(x)$ denotes the n -th derivative of $f(x) = \frac{1}{x}$, prove by mathematical induction that		3
$f^{(n)}(x) = \frac{(-1)^n n!}{x^{n+1}} \quad \forall n \in \mathbb{Z}^+$		
(c) i. Prove that $\forall a, b \in \mathbb{R}$,		1
$a^2 + b^2 \geq 2ab$		
ii. Prove that for $x, y \in \mathbb{R}^+$,		1
$\frac{x}{y} + \frac{y}{x} \geq 2$		
iii. Prove by induction, or otherwise, that		4
$(x_1 + x_2 + x_3 + \cdots + x_{n-1} + x_n) \left(\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \cdots + \frac{1}{x_{n-1}} + \frac{1}{x_n} \right) \geq n^2$		
where $x_i \in \mathbb{R}^+$, $i \in \mathbb{Z}^+$.		
(d) Consider the following statement:		2
$\forall a \in \mathbb{R}^+, \forall b \in \mathbb{R}^+, \log_{\frac{1}{a}} \frac{1}{b} = \log_a b$		
Either prove the statement is true, or give a counter example.		

Question 12 (14 Marks)

Commence a NEW booklet.

Marks

- (a) A particle is experiencing simple harmonic motion along a straight line. At time t seconds, its displacement x metres from a fixed point O on the line is given by

$$x = 6 \cos^2 t - 2$$

- i. Show that $\ddot{x} = -4(x - 1)$. **2**
 ii. Find the centre and period of the motion. **2**

- (b) The velocity of a particle moving in simple harmonic motion along the x axis is given by

$$v^2 = -x^2 - 4x + 12$$

- i. State the centre and period of the motion. **2**
 ii. What is the maximum speed of the particle? **1**

- (c) A particle moving in a straight line has displacement x metres from a fixed point O . Its acceleration is given by

$$\ddot{x} = \sqrt{3x + 4} \text{ ms}^{-2}$$

- i. Show that **1**

$$v^2 = \frac{4}{9} (3x + 4)^{\frac{3}{2}} + C$$

where v is the velocity of the particle in metres per second, and C is a constant.

- ii. Find the value of C if the particle commences from rest at $x = 0$. **1**
 iii. Explain why the motion of the particle is always in the positive direction. **1**

- (d) The velocity of a particle at time t seconds is given by **4**

$$\dot{\mathbf{r}}(t) = (4t - 3)\mathbf{i} + 2t\mathbf{j} - 5\mathbf{k}$$

where components are measured in metres per second.

Find the distance of the particle from the origin in metres when $t = 2$, given that $\mathbf{r}(0) = \mathbf{i} - 2\mathbf{j}$.

Examination continues overleaf...

Question 13 (14 Marks)

Commence a NEW booklet.

Marks

- (a) The equations of lines ℓ_1 and ℓ_2 are given with respect to a fixed origin: **2**

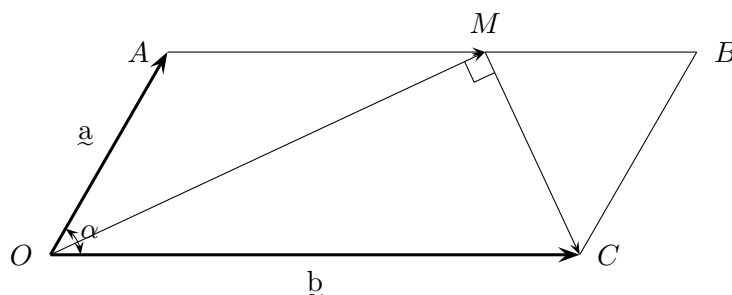
$$\ell_1 : \underline{r}_1 = \begin{pmatrix} 11 \\ 2 \\ 17 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix}$$

$$\ell_2 : \underline{r}_2 = \begin{pmatrix} -5 \\ 11 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} p \\ 2 \\ 2 \end{pmatrix}$$

where λ and μ are parameters, and p is a constant.

What is the value of p if $\ell_1 \perp \ell_2$?

- (b) The diagram below shows the parallelogram $OABC$ where $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OC} = \underline{b}$ and $|\overrightarrow{OC}| = 2|\overrightarrow{OA}|$. The angle between $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is α .



M is a point on AB such that $\overrightarrow{AM} = k\overrightarrow{AB}$, $k \in [0, 1]$ and $OM \perp MC$.

- i. Use a vector method to show that **4**

$$|\underline{a}|^2 (1 - 2k)(2 \cos \alpha - (1 - 2k)) = 0$$

- ii. Find the possible values of α for point M to satisfy the given conditions. **2**

Examination continues overleaf...

Question 13 continued from the previous page...

- (c) A ball is projected in the uphill direction from the base of the hill with velocity vector

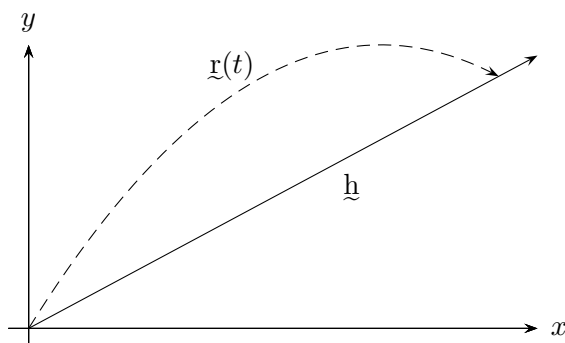
$$\dot{\mathbf{r}}(t) = \begin{pmatrix} 10 \\ 10\sqrt{3} - 10t \end{pmatrix} \text{ ms}^{-1}$$

where t is measured in seconds.

The hill is represented by the line with vector equation

$$\underline{\mathbf{h}} = \lambda \begin{pmatrix} 10\sqrt{3} \\ 10 \end{pmatrix} \text{ metres}$$

where $\lambda \in \mathbb{R}^+$



- i. Show that the Cartesian equation of the trajectory is

2

$$y = x\sqrt{3} - \frac{x^2}{20}$$

- ii. Hence find the coordinates where the ball will land along the hill.

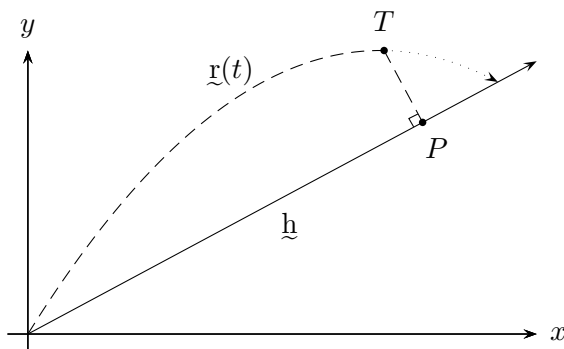
2

- iii. Find the instantaneous velocity of the shadow which the ball casts on the uphill slope, 2 seconds after the ball has been launched, in the form

2

$$k \begin{pmatrix} a \\ b \end{pmatrix} \text{ metres per second}$$

For simplicity, you may assume that the direction of the sunlight is perpendicular to the hill.



Examination continues overleaf...

Question 14 (18 Marks)

Commence a NEW booklet.

Marks

- (a) The polynomial $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$ has roots $a + bi$ and $a + 2bi$ where a and $b \in \mathbb{R}$ and $b \neq 0$. **3**

By evaluating a and b , find all the roots of $P(x)$.

- (b) i. If $z = \cos \theta + i \sin \theta$, show that **2**

$$z^n + z^{-n} = 2 \cos n\theta$$

- ii. If $z + \frac{1}{z} = u$, find an expression for $z^3 + \frac{1}{z^3}$ in terms of u . **2**

- iii. It can be shown that $z^5 + \frac{1}{z^5} = u^5 - 5u^3 + 5u$ (Do NOT prove this). **3**

Show that

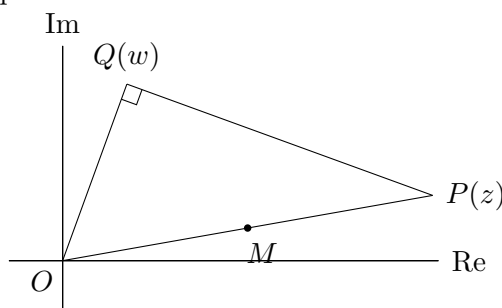
$$1 + \cos(10\theta) = 2(16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta)^2$$

- (c) The point $P(x, y)$ representing the complex number moves in the Argand diagram so that $|z - 6| = |z + 2i|$.

- i. Show that the path P traces out, has equation $3x + y - 8 = 0$. **2**

- ii. Find the minimum value of $|z|$ as P moves on this path. **2**

- (d) In the Argand diagram below, $\triangle OPQ$ is right-angled at Q and $QP = kOQ$ for $k \in \mathbb{R}^+$. M is the midpoint of OP .



- i. If the complex number z is represented by the vector $\overrightarrow{OP}$ and the complex number w is represented by $\overrightarrow{OQ}$, show that **2**

$$\overrightarrow{OM} = \frac{1}{2}(1 - ki)w$$

- ii. Express $\overrightarrow{MQ}$ in terms of w , and hence show that **2**

$$|\overrightarrow{OM}| = |\overrightarrow{MQ}|$$

Examination continues overleaf...

Question 15 (14 Marks)

Commence a NEW booklet.

Marks

- (a) Using the substitution $u = 1 - \sin 2x$, evaluate 4

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sqrt{1 - \sin 2x} (1 - 2 \cos^2 x) dx$$

- (b) i. Find the value of A , B and C such that 1

$$\frac{1}{x(1+x^2)} \equiv \frac{A}{x} + \frac{Bx+C}{1+x^2}$$

- ii. Hence evaluate 2

$$\int_1^{\sqrt{3}} \frac{1}{x(1+x^2)} dx$$

Express your answer in the form $\log_e \left(\sqrt{\frac{a}{b}} \right)$ where $a, b \in \mathbb{R}^+$.

- (c) Let $I_n = \int_0^1 x^n \tan^{-1} x dx$ where $n = 0, 1, 2, \dots$

- i. Show that 2

$$(n+1)I_n = \frac{\pi}{4} - \int_0^1 \frac{x^{n+1}}{1+x^2} dx$$

for $n \geq 0$.

- ii. Hence or otherwise, find the value of I_0 . 1

- iii. Show that $(n+3)I_{n+2} + (n+1)I_n = \frac{\pi}{2} - \frac{1}{n+2}$ 2

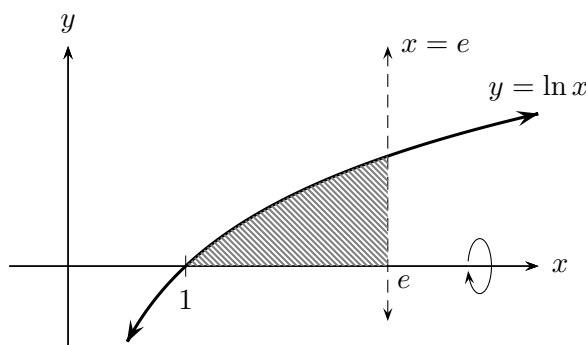
- iv. Hence find the value of I_4 . 2

Question 16 (17 Marks)

Commence a NEW booklet.

Marks

- (a) The region enclosed by the curve $y = \ln x$, the line $x = e$ and the x axis is shown in the diagram. 3



Find the volume of the solid formed by rotating the shaded region about the x axis.

Examination continues overleaf...

Question 16 continued from previous page...

- (b) Sphere S has vector equation

$$|\underline{r} - (3\underline{i} + \underline{j} + 4\underline{k})| = \sqrt{35}$$

- i. Write the Cartesian equation of this sphere. 1

- ii. The line ℓ has equation $\underline{r} = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$, $\lambda \in \mathbb{R}$. 3

Determine whether this line is a tangent to the sphere S or not. Give full reasons for your answer.

- (c) A car of mass m kg is being driven along a straight road. The engine of the car provides a constant propelling force mP , $P \in \mathbb{R}^+$, while the car experiences a resistive force of mkv^2 , where v ms⁻¹ is the velocity of the car, and k is a positive constant. The car is initially at rest.

- i. By drawing a diagram or otherwise, show that the situation described can be modelled by a differential equation of the form 1

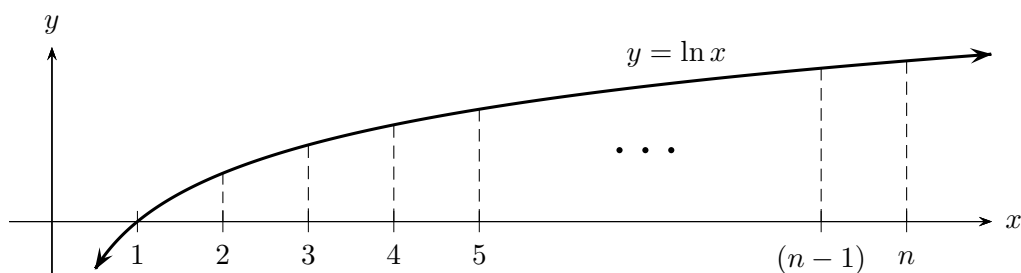
$$\frac{dv}{dx} = g(v)$$

where $g(v)$ is a function of v , and x is the displacement of the car from its initial position in metres.

- ii. Show that $v^2 = \frac{P}{k}(1 - e^{-2kx})$. Hence or otherwise, explain why the maximum speed of the car is $v_M = \sqrt{\frac{P}{k}}$ ms⁻¹. 3

- iii. Show that the distance required for the car to reach a speed of $\frac{1}{3}v_M$ is approximately 41% of the distance required to reach a speed of $\frac{1}{2}v_M$. 1

- (d) The following diagram shows the graph of $y = \ln x$ and $(n - 1)$ strips of equal width from $x = 1$ to $x = n$.



- i. Show that 2

$$\frac{\ln 1 + \ln 2}{2} + \frac{\ln 2 + \ln 3}{2} + \cdots + \frac{\ln(n-1) + \ln n}{2} < \int_1^n \ln x \, dx$$

- ii. Hence deduce that $n! < \frac{en^{n+\frac{1}{2}}}{e^n}$ 3

End of paper.

Sample Band E4 Responses

Section I

1. (A) 2. (B) 3. (A) 4. (A) 5. (A)
6. (C) 7. (D) 8. (C) 9. (A) 10. (D)

Section II

Question 11 (Ham)

(a) (2 marks)

✓ [1] for rewriting as $100x + 10y + z$.

✓ [1] for using the given fact and undeniably proving $n = 11R$

Let the number be $n = 100x + 10y + z$, such that $x + z = y$.

$$\begin{aligned} n &= 100x + 10(x + z) + z \\ &= 100x + 10x + 10z + z \\ &= 110x + 11z \\ &= 11(10x + z) = 11R \end{aligned}$$

such that $R \in \mathbb{Z}^+$. Hence n is divisible by 11.

(b) (3 marks)

✓ [1] for proving the base case.

✓ [1] for the inductive hypothesis, and for progress in differentiating.

✓ [1] for final answer.

Let $P(n)$ be the proposition

$$f^{(n)}(x) = \frac{(-1)^n n!}{x^{n+1}} \quad \forall n \in \mathbb{Z}^+$$

such that $f^{(n)}(x)$ denotes the n -th derivative of $f(x) = \frac{1}{x}$.

- Base case: $P(1)$:

$$\begin{aligned} \text{LHS} \quad \frac{d}{dx}(x^{-1}) &= -x^{-2} = -\frac{1}{x^2} \\ \text{RHS} \quad f^{(1)}(x) &= \frac{(-1)^1 \times 1!}{x^{1+1}} = \frac{-1}{x^2} \end{aligned}$$

Hence $P(1)$ is true.

- Inductive step: assume $P(k)$ is true, $k \in \mathbb{Z}^+$:

$$f^{(k)}(x) = \frac{(-1)^k k!}{x^{k+1}} \quad \forall k \in \mathbb{Z}^+$$

such that $f^{(k)}(x)$ denotes the k -th derivative of $f(x) = \frac{1}{x}$.

Examine $P(k+1)$:

$$\begin{aligned}\frac{d}{dx} \left(f^k(x) \right) &= \frac{d}{dx} \left((-1)^k k! \left(x^{-(k+1)} \right) \right) \\ &= (-1)^{k+1} k! \times (k+1) x^{-(k+1)-1} \\ &= \frac{(-1)^{k+1} (k+1)!}{x^{k+2}} \\ &= \frac{(-1)^{k+1} (k+1)!}{x^{(k+1)+1}}\end{aligned}$$

(c) i. (1 mark)

$$\begin{aligned}(a-b)^2 &\geq 0 \\ a^2 - 2ab + b^2 &\geq 0 \\ &\quad +2ab \quad +2ab \\ \therefore a^2 + b^2 &\geq 2ab\end{aligned}\tag{11.1}$$

ii. (1 mark) - From (11.1), if $b = \frac{1}{a}$,

$$\begin{aligned}a^2 + \frac{1}{a^2} &\geq 2a \times \frac{1}{2} \\ \therefore a^2 + \frac{1}{a^2} &\geq 2\end{aligned}$$

As $a^2 \geq 0$, and $x \in \mathbb{R}^+$ and $y \in \mathbb{R}^+$, replace a^2 with $\frac{x}{y}$ and $\frac{1}{a^2}$ with $\frac{y}{x}$:

$$\therefore \frac{x}{y} + \frac{y}{x} \geq 2$$

iii. (4 marks)

- ✓ [1] for proving the base case.
- ✓ [1] for use of $X \times Y \geq k^2$
- ✓ [1] for correct summation after expansion of $\frac{X}{x_{k+1}} + Yx_{k+1}$
- ✓ [1] for final answer

Let $P(n)$ be the proposition:

$$(x_1 + x_2 + x_3 + \cdots + x_{n-1} + x_n) \left(\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \cdots + \frac{1}{x_{n-1}} + \frac{1}{x_n} \right) \geq n^2$$

where $x_i \in \mathbb{R}^+$, $i \in \mathbb{Z}^+$.

- Base case: $P(1)$:

$$\begin{array}{ll}\text{LHS} & x_1 \times \frac{1}{x_1} = 1 \\ \text{RHS} & 1^2 \geq 1\end{array}$$

Hence $P(1)$ is true.

- Inductive hypothesis: assume $P(k)$ is true, i.e.

$$(x_1 + x_2 + x_3 + \cdots + x_{k-1} + x_k) \left(\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \cdots + \frac{1}{x_{k-1}} + \frac{1}{x_k} \right) \geq k^2$$

where $x_i \in \mathbb{R}^+$, $i \in \mathbb{Z}^+$.

Examine $P(k+1)$:

$$\begin{aligned} & (x_1 + x_2 + x_3 + \cdots + x_{k-1} + x_k + \textcolor{red}{x_{k+1}}) \left(\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \cdots + \frac{1}{x_{k-1}} + \frac{1}{x_k} + \frac{\textcolor{red}{1}}{\textcolor{red}{x_{k+1}}} \right) \\ & \equiv (X + x_{k+1}) \left(Y + \frac{1}{x_{k+1}} \right) \end{aligned}$$

Where $X = x_1 + x_2 + x_3 + \cdots + x_{k-1} + x_k$ and $Y = \frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \cdots + \frac{1}{x_{k-1}} + \frac{1}{x_k}$

$$\begin{aligned} & = XY + \frac{X}{x_{k+1}} + Yx_{k+1} + \frac{x_{k+1}}{x_{k+1}} \\ & \geq \overbrace{k^2}^{P(k)} + 1 + \left(\frac{x_1}{x_{k+1}} + \frac{x_{k+1}}{x_1} + \frac{\textcolor{red}{x_2}}{\textcolor{red}{x_{k+1}}} + \frac{\textcolor{red}{x_{k+1}}}{\textcolor{red}{x_2}} + \frac{\textcolor{blue}{x_3}}{\textcolor{blue}{x_{k+1}}} + \frac{\textcolor{blue}{x_{k+1}}}{\textcolor{blue}{x_3}} + \cdots + \frac{x_k}{x_{k+1}} + \frac{x_{k+1}}{x_k} \right) \\ & \geq k^2 + 1 + \underbrace{(2 + \textcolor{red}{2} + \textcolor{blue}{2} + \cdots + 2)}_{k \text{ times}} \\ & = k^2 + 2k + 1 \\ & = (k+1)^2 \end{aligned}$$

(d) (2 marks)

- ✓ [1] for using the change of base rule
- ✓ [1] for giving a reason why the statement is false.
- ✓ [2] if student immediately realises the problem with $a = 1$ and concludes it's a false statement.

$$\begin{aligned} & \forall a \in \mathbb{R}^+, \forall b \in \mathbb{R}^+ \\ & \log_{\frac{1}{a}} \frac{1}{b} = \frac{\log_a \frac{1}{b}}{\log_a \frac{1}{a}} \quad (\text{Change of base rule}) \\ & = \frac{\log_a (b^{-1})}{\log_a (a^{-1})} \\ & = \frac{-\log_a b}{-\log_a a} \\ & = \log_a b \end{aligned}$$

However if $a = 1$, then

$$\begin{aligned} & \log_{\frac{1}{1}} \frac{1}{b} \Rightarrow \frac{1}{b} = 1 \\ & \therefore b = 1 \text{ only} \end{aligned}$$

So when $a = 1$, the statement is not true for all $b \in \mathbb{R}^+$, as only one value of b fits.

- Hence the statement is true $\forall a \in \mathbb{R}^+$ and $\forall b \in \mathbb{R}^+$, except $a = 1$ (which locks in $b = 1$ only and not $\forall b \in \mathbb{R}^+$)

Question 12 (Ham)

(c) i. (1 mark)

(a) i. (2 marks)

✓ [1] for finding the displacement in terms of $\cos 2t$

✓ [1] for showing required result.

$$\begin{aligned}
 x &= 6 \cos^2 t - 2 \\
 &= 6 \left(\frac{1}{2} + \frac{1}{2} \cos 2t \right) - 2 \\
 &= 3 + 3 \cos 2t - 2 \\
 &= 1 + 3 \cos 2t \\
 x - 1 &= 3 \cos 2t
 \end{aligned}$$

Differentiating w.r.t. t :

$$\dot{x} = -3(2) \sin 2t$$

Differentiate again,

$$\begin{aligned}
 \ddot{x} &= -3(2^2) \cos 2t \\
 &= -2^2(x - 1) = -4(x - 1)
 \end{aligned}$$

ii. (2 marks)

✓ [1] for each of the correct *centre of motion* and *period*.

$$c = 1$$

$$T = \frac{2\pi}{2} = \pi \text{ seconds}$$

(b) i. (2 marks)

✓ [1] for each of the correct *centre of motion* and *period*.

$$\begin{aligned}
 v^2 &= -x^2 - 4x + 12 \\
 &= -(x^2 + 4x + 4) + 16 \\
 &= 16 - (x + 2)^2 \\
 &= 1 \left(4^2 - (x + 2)^2 \right) \\
 &\equiv n^2 \left(a^2 - (x - x_0)^2 \right) \\
 \therefore a &= 4 \quad x_0 = -2 \\
 T &= \frac{2\pi}{1} = 2\pi \text{ seconds}
 \end{aligned}$$

ii. (1 mark)

$$v_{\max} = \sqrt{n^2(a^2 - 0)} = 4$$

$$\ddot{x} = (3x + 4)^{\frac{1}{2}}$$

$$\begin{aligned}
 \frac{1}{2}v^2 &= \int \ddot{x} dx \\
 &= \int (3x + 4)^{\frac{1}{2}} dx \\
 &= \frac{(3x + 4)^{\frac{3}{2}}}{\frac{3}{2} \times 3} + C_1 \\
 &= \frac{2(3x + 4)^{\frac{3}{2}}}{9} + C_1 \\
 \therefore v^2 &= \frac{4}{9}(3x + 4)^{\frac{3}{2}} + C_1
 \end{aligned}$$

ii. (1 mark)

When $t = 0$, $x = 0$ and $v = 0$:

$$\begin{aligned}
 0 &= \frac{4}{9}(0 + 4)^{\frac{3}{2}} + C_1 \\
 0 &= \frac{4}{9} \times 8 + C_1 \\
 C_1 &= -\frac{32}{9} \\
 v^2 &= \frac{4}{9}(3x + 4)^{\frac{3}{2}} - \frac{32}{9}
 \end{aligned}$$

iii. (1 mark)

$$\begin{aligned}
 \ddot{x} &= \sqrt{3x + 4} \Big|_{x=0} \\
 &= \sqrt{0 + 4} = 2 \text{ ms}^{-2}
 \end{aligned}$$

As $v = 0$ when $t = 0$ and $\ddot{x} > 0$ the particle will move in the positive x direction. Hence,

$$v = +\sqrt{\frac{4}{9}(3x + 4)^{\frac{3}{2}} - \frac{32}{9}} > 0$$

In fact the particle will always move in the positive direction given the sign of v .

(d) (4 marks)

- ✓ [1] for each correct arbitrary constant of integration given the initial conditions.
- ✓ [1] for distance from the origin.

$$\dot{\mathbf{r}}(t) = (4t - 3)\mathbf{i} + (2t)\mathbf{j} - 5\mathbf{k}$$

Integrating the respective components,

$$\mathbf{r}(t) = \begin{pmatrix} 2t^2 - 3t + C_1 \\ t^2 + C_2 \\ -5t + C_3 \end{pmatrix}$$

$$\text{As } \mathbf{r}(0) = \mathbf{i} - 2\mathbf{j},$$

$$\begin{pmatrix} (0 + C_1) \\ (0 + C_2) \\ -(0 + C_3) \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$$

$$\therefore C_1 = 1 \quad C_2 = -2 \quad C_3 = 0$$

$$\mathbf{r}(t) = \begin{pmatrix} 2t^2 - 3t + 1 \\ t^2 - 2 \\ -5t \end{pmatrix}$$

When $t = 2$,

$$\mathbf{r}(2) = \begin{pmatrix} (8 - 6 + 1) \\ 2 \\ -10 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -10 \end{pmatrix}$$

At $t = 2$, the particle's distance from the origin is

$$d = \sqrt{3^2 + 2^2 + (-10)^2} = \sqrt{113}$$

Question 13 (Ham)

(a) (2 marks)

- ✓ [1] for applying the dot product to the vector components (excluding the fixed point), and setting to zero for the lines to be perpendicular.
- ✓ [1] for finding the correct value of p .

$$\ell_1 : \mathbf{r}_1 = \begin{pmatrix} 11 \\ 2 \\ 17 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix}$$

$$\ell_2 : \mathbf{r}_2 = \begin{pmatrix} -5 \\ 11 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} p \\ 2 \\ 2 \end{pmatrix}$$

When $\ell_1 \perp \ell_2$, the parallel vector components of ℓ_1 and ℓ_2 are perpendicular:

$$\lambda \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix} \cdot \mu \begin{pmatrix} p \\ 2 \\ 2 \end{pmatrix} = 0$$

$$\begin{pmatrix} -2\lambda \\ \lambda \\ -4\lambda \end{pmatrix} \cdot \begin{pmatrix} \mu p \\ 2\mu \\ 2\mu \end{pmatrix} = 0$$

$$-2\lambda\mu p + 2\lambda\mu - 8\lambda\mu = 0$$

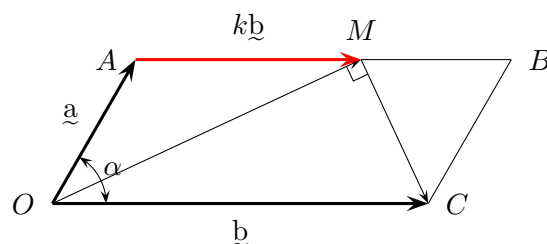
$$-2p + 2 - 8 = 0$$

$$-6 = 2p$$

$$p = -3$$

(b) i. (4 marks)

- ✓ [1] for obtaining $\overrightarrow{MC}$ in terms of $\mathbf{a}$, $\mathbf{b}$ and k
- ✓ [1] for observing $\mathbf{a} \cdot \mathbf{b} = 2|\mathbf{a}|^2 \cos \alpha$
- ✓ [1] for substantial further working to calculate $\overrightarrow{OM} \cdot \overrightarrow{MC} = 0$.
- ✓ [1] for final result required.



By vector addition,

$$\overrightarrow{OM} + \overrightarrow{MC} = \overrightarrow{OC}$$

$$\overbrace{\mathbf{a} + k\mathbf{b}}^{\overrightarrow{OM}} + \overrightarrow{MC} = \mathbf{b}$$

$$\therefore \overrightarrow{MC} = \mathbf{b} - \mathbf{a} - k\mathbf{b}$$

$$= \mathbf{b}(1 - k) - \mathbf{a}$$

Some additional facts:

- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$
- $|\overrightarrow{OC}| = 2|\overrightarrow{OA}|$, i.e. $|\mathbf{b}| = 2|\mathbf{a}|$

$$\therefore \mathbf{b} \cdot \mathbf{b} = |\mathbf{b}|^2 = 4|\mathbf{a}|^2$$

- $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \alpha = 2|\mathbf{a}|^2 \cos \alpha$

As $\overrightarrow{OM} \cdot \overrightarrow{MC} = 0$, and $\overrightarrow{OM} = \underline{a} + k\underline{b}$, (c) i. (2 marks)

$$\begin{aligned} 0 &= \overbrace{(\underline{a} + k\underline{b})}^{\overrightarrow{OM}} \cdot \overbrace{(\underline{b}(1-k) - \underline{a})}^{\overrightarrow{MC}} \\ &= \underline{a} \cdot \underline{b}(1-k) - (\underline{a} \cdot \underline{a}) + k(\underline{b} \cdot \underline{b})(1-k) \\ &\quad - (k\underline{a} \cdot \underline{b}) \\ &= 2(1-k)|\underline{a}|^2 \cos \alpha - (|\underline{a}|^2) \\ &\quad + k(4|\underline{a}|^2)(1-k) - 2k|\underline{a}|^2 \cos \alpha \\ &= (2-4k)|\underline{a}|^2 \cos \alpha + |\underline{a}|^2(4k(1-k) - 1) \\ &= (2-4k)|\underline{a}|^2 \cos \alpha + |\underline{a}|^2(4k - 4k^2 - 1) \\ &= 2(1-2k)|\underline{a}|^2 \cos \alpha - |\underline{a}|^2(1-2k)^2 \\ &= |\underline{a}|^2(1-2k)[2 \cos \alpha - (1-2k)] \end{aligned}$$

ii. (2 marks)

✓ [1] for each correct boundary.

✓ [1] for correct inequality.

From above,

$$|\underline{a}|^2(1-2k)[2 \cos \alpha - (1-2k)] = 0$$

Using the null factor law,

$$\begin{aligned} (1-2k) &= 0 \\ k &= \frac{1}{2} \end{aligned}$$

Finding the other option,

$$\begin{aligned} 2 \cos \alpha - (1-2k) &= 0 \quad k \in [0, 1] \\ 2k &= 1 - 2 \cos \alpha \\ k &= \frac{1}{2} - \cos \alpha \end{aligned}$$

As $k \in [0, 1]$,

$$\begin{aligned} 0 &\leq \underbrace{\frac{1}{2} - \cos \alpha}_{\times(-1)} \leq 1 \\ -1 &\leq \cos \alpha - \frac{1}{2} \leq 0 \\ -\frac{1}{2} &\leq \cos \alpha \leq \frac{1}{2} \end{aligned}$$

As $\alpha > 0$,

$$\therefore \frac{\pi}{3} \leq \alpha \leq \frac{2\pi}{3}$$

✓ [1] for correct integration to obtain $\underline{i}$ and $\underline{j}$ components.

✓ [1] for eliminating the t parameter to obtain the Cartesian equation.

$$\begin{aligned} \dot{\underline{x}}(t) &= \left(\frac{10}{10\sqrt{3} - 10t} \right) \text{ ms}^{-1} \\ \dot{x} &= 10 & \dot{y} &= 10\sqrt{3} - 10t \\ x &= \int \dot{x} dt & y &= \int \dot{y} dt \\ &= \int 10 dt & &= \int (10\sqrt{3} - 10t) dt \\ &= 10t + C_1 & &= 10t\sqrt{3} - 5t^2 + C_2 \end{aligned}$$

When $t = 0$, $x = 0$ and $y = 0$:

$$\therefore x = 10t \quad y = 10t\sqrt{3} - 5t^2$$

Change subject in horizontal displacement equation:

$$t = \frac{x}{10}$$

Substitute into vertical displacement equation:

$$\begin{aligned} y &= 10 \left(\frac{x}{10} \right) \sqrt{3} - 5 \left(\frac{x}{10} \right)^2 \\ &= x\sqrt{3} - \frac{x^2}{20} \end{aligned}$$

ii. (2 marks)

✓ [1] for each correct coordinate

$$\begin{aligned} \underline{h} &= \lambda \left(\frac{10\sqrt{3}}{10} \right) \\ \therefore x &= 10\sqrt{3} & y &= 10 \\ m &= \frac{10}{10\sqrt{3}} = \frac{1}{\sqrt{3}} \end{aligned}$$

Using $y = mx + c$ where $c = 0$ as the projectile commences from the origin, the hill has equation

$$y = \frac{1}{\sqrt{3}}x$$

Solving simultaneously with the Cartesian path of the projectile,

$$\begin{aligned} \begin{cases} y = x\sqrt{3} - \frac{x^2}{20} \\ y = \frac{1}{\sqrt{3}}x \end{cases} \\ \frac{1}{\sqrt{3}}x = x\sqrt{3} - \frac{x^2}{20} \\ x\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) - \frac{1}{20}x^2 = 0 \\ x\left(\left(\frac{3}{\sqrt{3}} - \frac{1}{\sqrt{3}}\right) - \frac{x}{20}\right) = 0 \\ x = 20\left(\frac{2}{\sqrt{3}}\right) = \frac{40}{\sqrt{3}} \\ y = \frac{x}{\sqrt{3}} = \frac{40}{3} \end{aligned}$$

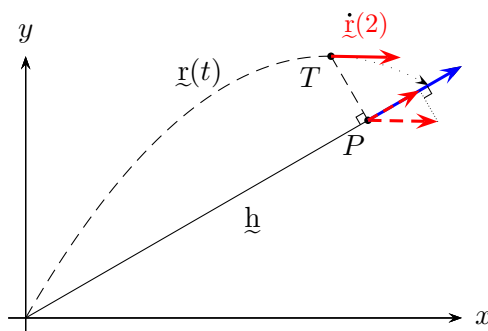
to $\underline{v}$, such that

$$\begin{aligned} \underline{u} = \dot{\underline{r}}(2) &= \begin{pmatrix} 10 \\ 10\sqrt{3} - 20 \end{pmatrix} \\ \underline{v} &= \begin{pmatrix} 10\sqrt{3} \\ 10 \end{pmatrix} \\ \text{proj}_{\underline{v}} \underline{u} &= \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \underline{v} \\ &= \frac{10(10\sqrt{3}) + 10(10\sqrt{3} - 20)}{300 + 100} \begin{pmatrix} 10\sqrt{3} \\ 10 \end{pmatrix} \\ &= \frac{200\sqrt{3} - 200}{400} \begin{pmatrix} 10\sqrt{3} \\ 10 \end{pmatrix} \\ &= (\sqrt{3} - 1) \begin{pmatrix} 5\sqrt{3} \\ 5 \end{pmatrix} \end{aligned}$$

iii. (2 marks)

- ✓ [1] for correctly finding $\dot{\underline{r}}(2)$.
- ✓ [1] for correctly applying the projection formula and finding the velocity of the shadow along the hill.
- The velocity of the shadow cast by the ball in the uphill direction is the projection of the velocity of the ball at $t = 2$ on to the hill.

$$\dot{\underline{r}}(2) = \begin{pmatrix} 10 \\ 10\sqrt{3} - 20 \end{pmatrix}$$



- At $t = 2$,

$$\dot{\underline{r}}(2) = \begin{pmatrix} 10 \\ 10\sqrt{3} - 20 \end{pmatrix}$$

- Applying the projection of $\underline{u}$ on

Question 14 (Lam)

(a) (3 marks)

- ✓ [1] for finding both complex conjugate roots.
- ✓ [1] for value of a .
- ✓ [1] for value of b .

$$x^4 - 4x^3 + 11x^2 - 14x + 10 = 0$$

As $P(x)$ has real coefficients, then any complex roots that appear will also have its conjugate appear as a root. Hence,

- $a + bi$ and $a - bi$
- $a + 2bi$ and $a - 2bi$

are all roots. Examine the sum of roots:

$$\begin{aligned} (a + bi) + (a - bi) + (a + 2bi) + (a - 2bi) &= -\frac{b}{a} \\ 4a &= -\frac{-4}{1} = 4 \\ \therefore a &= 1 \end{aligned}$$

Examine the product of roots,

$$\begin{aligned} (a + bi)(a - bi)(a + 2bi)(a - 2bi) &= \frac{e}{a} \\ (a^2 + b^2)(a^2 + 4b^2) &= 10 \end{aligned}$$

As $a = 1$,

$$\begin{aligned}(1 + b^2)(1 + 4b^2) &= 10 \\ 1 + 4b^2 + b^2 + 4b^4 &= 10 \\ 4b^4 + 5b^2 - 9 &= 0 \\ (4b^2 + 9)(b^2 - 1) &= 0 \\ \therefore b &= \pm 1\end{aligned}$$

Hence roots are $1 \pm i$ and $1 \pm 2i$.

(b) i. (2 marks)

- ✓ [1] for usage of De Moivre's Theorem to obtain z^n .
- ✓ [1] for showing the final result.

$$z = \cos \theta + i \sin \theta$$

By De Moivre's Theorem,

$$\begin{aligned}z^n &= \cos n\theta + i \sin n\theta \\ z^{-n} &= \cos(-n\theta) + i \sin(-n\theta) \\ &= \cos(n\theta) - i \sin(n\theta)\end{aligned}$$

Due to $\cos$ being an even function and $\sin$ being an odd function.

$$\begin{aligned}\therefore z^n + z^{-n} &= (\cos n\theta + i \sin n\theta) + \\ &\quad (\cos(n\theta) - i \sin(n\theta)) \\ &= 2 \cos n\theta\end{aligned}$$

ii. (2 marks)

- ✓ [1] for expanding $(z + \frac{1}{z})^3$ via the binomial theorem.
- ✓ [1] for showing the final result.

$$\begin{aligned}\text{Let } u &= z^1 + \frac{1}{z^1} \\ \left(z + \frac{1}{z}\right)^3 & \\ &= z^3 + 3z^2 \left(\frac{1}{z}\right) + 3z \left(\frac{1}{z^2}\right) + \frac{1}{z^3} \\ &= z^3 + 3z + \frac{3}{z} + \frac{1}{z^3} \\ &= z^3 + \frac{1}{z^3} + 3 \left(z + \frac{1}{z}\right) \\ \therefore u^3 &= z^3 + \frac{1}{z^3} + 3u \\ z^3 + \frac{1}{z^3} &= u^3 - 3u\end{aligned}$$

iii. (3 marks)

- ✓ [1] for using $u = z + \frac{1}{z}$ in the given expression.
- ✓ [1] for using the double angle formula for $\cos 10\theta$.
- ✓ [1] for final result.

$$\begin{aligned}z^5 + \frac{1}{z^5} &= \left(z + \frac{1}{z}\right)^5 - 5 \left(z + \frac{1}{z}\right)^3 + 5 \left(z + \frac{1}{z}\right) \\ &= (2 \cos \theta)^5 - 5 (2 \cos \theta)^3 + 5 (2 \cos \theta) \\ &= 32 \cos^5 \theta - 50 \cos^3 \theta + 10 \cos \theta \\ &= 2 \cos 5\theta \quad \text{from (i)}\end{aligned}$$

Hence,

$$\begin{aligned}\cos 5\theta &= \frac{1}{2} (\cos^5 \theta - 40 \cos^3 \theta + 10 \cos \theta) \\ &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta\end{aligned}$$

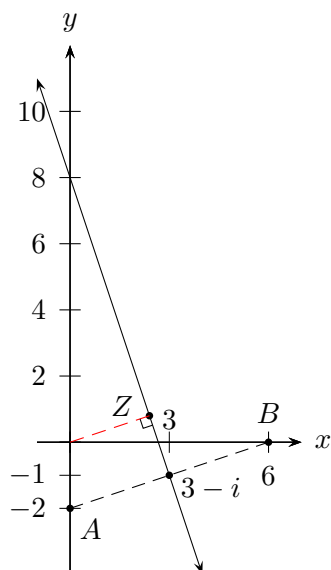
Also, $\cos 10\theta = 2 \cos^2 5\theta - 1$:

$$\begin{aligned}\therefore 1 + \cos 10\theta &= 2 \cos^2 5\theta \\ &= 2 (16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta)^2\end{aligned}$$

(c) i. (2 marks)

- ✓ [1] for identifying the path being the perpendicular bisector of the interval.
- ✓ [1] finding the equation of the path traced out.

$|z - 6| = |z + 2i|$ is the perpendicular bisector of the interval between $6 + 0i$ and $0 - 2i$.



$$MP = \left(\frac{6}{2}, -1\right) = (3, -1)$$

$$m = \frac{2}{6} = \frac{1}{3}$$

$$\therefore m_{\perp} = -3$$

Applying the point-gradient formula, (d)

$$y + 1 = -3(x - 3)$$

$$y + 1 = -3x + 9$$

$$y = -3x + 8$$

$$\therefore 3x + y - 8 = 0$$

ii. (2 marks)

- ✓ [1] for finding the equation from O to Z .
- ✓ [1] for finding the $|z|_{\min}$
- $|z|_{\min}$ when z is at the point where the perpendicular distance from the origin coincides.
- OZ is parallel to the interval AB , and hence

$$y = \frac{1}{3}x$$

is the equation of OZ .

- Find the point of intersection of OZ and the path traced out by P :

$$\begin{cases} y = -3x + 8 \\ y = \frac{1}{3}x \end{cases}$$

$$\frac{1}{3}x = -3x + 8$$

$$\frac{10}{3}x = 8$$

$$\therefore x = \frac{24}{10} = \frac{12}{5}$$

$$y = \frac{1}{3} \left(\frac{12}{5}\right) = \frac{4}{5}$$

Apply the distance formula,

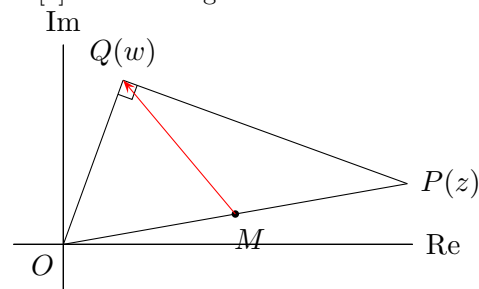
$$|z| = \sqrt{\left(\frac{12}{5}\right)^2 + \left(\frac{4}{5}\right)^2}$$

$$= \sqrt{\frac{160}{25}}$$

$$= \frac{4}{5}\sqrt{10}$$

i. (2 marks)

- ✓ [1] for substantial progress in the working.
- ✓ [1] for showing the final result.



$$\overrightarrow{OM} = \frac{1}{2}z \text{ (midpoint)}$$

$$\overrightarrow{OQ} + \overrightarrow{QP} = \overrightarrow{OP}$$

$$w + \overrightarrow{QP} = z$$

Also, $QP = kOQ$:

$$\overrightarrow{QP} = z - w$$

$$= -iwk$$

as QP is a rotation of the vector w by -90° . Hence, **Question 15** (Lam)

(a) (4 marks)

$$z = w - iwk = w(1 - ki)$$

$$\therefore \overrightarrow{OM} = \frac{1}{2}z = \frac{1}{2}w(1 - ki)$$

ii. (2 marks)

✓ [1] for finding $\overrightarrow{MQ}$ in terms of w and k .

✓ [1] for showing final result, which must include the correct application of the modulus.

✓ [1] for transforming the differential.

✓ [1] for transforming both limits.

✓ [1] for transforming integrand to an integrable form.

✓ [1] for final answer.

$$u = 1 - \sin 2x$$

$$\therefore du = -2 \cos 2x \, dx$$

Transforming the limits,

$$\overrightarrow{OM} + \overrightarrow{MQ} = \overrightarrow{OQ}$$

$$\frac{1}{2}w(1 - ki) + \overrightarrow{MQ} = w$$

$$\overrightarrow{MQ} = w - \frac{1}{2}w(1 - ki)$$

$$= w - \frac{1}{2}w + \frac{1}{2}kiw$$

$$= \frac{1}{2}w(1 + ki)$$

$$|\overrightarrow{MQ}| = \left| \frac{1}{2} \right| |w| |1 + ki|$$

$$= \left| \frac{1}{2} \right| |w| \sqrt{1 + k^2}$$

Also, the magnitude of $\overrightarrow{OM}$:

$$|\overrightarrow{OM}| = \left| \frac{1}{2}w(1 - ki) \right|$$

$$= \left| \frac{1}{2} \right| |w| |1 - ki|$$

$$= \left| \frac{1}{2} \right| |w| \sqrt{1 + (-k)^2}$$

$$= \left| \frac{1}{2} \right| |w| \sqrt{1 + k^2}$$

$$\therefore |\overrightarrow{MQ}| = |\overrightarrow{OM}|$$

$$x = \frac{\pi}{4} \quad u = 1 - \sin \frac{\pi}{2} = 0$$

$$x = \frac{\pi}{2} \quad u = 1 - \sin \pi = 1$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sqrt{1 - \sin 2x} (1 - 2 \cos^2 x) \, dx$$

$$= \int_{u=0}^{u=1} \sqrt{u} (-\cos 2x) \, dx$$

$$= \frac{1}{2} \int_{u=0}^{u=1} \sqrt{u} \overbrace{(-2 \cos 2x) \, dx}^{=du}$$

$$= \frac{1}{2} \int_0^1 \sqrt{u} \, du$$

$$= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_0^1$$

$$= \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$$

(b) i. (1 mark)

$$A = 1 \quad B = -1 \quad C = 0$$

(For brevity, only the answers are shown - use method of undetermined coefficients to find A , B and C).

ii. (2 marks)

- ✓ [1] for finding the primitive.
- ✓ [1] for full simplification to the required form.

$$\begin{aligned}
 & \int_1^{\sqrt{3}} \frac{1}{x(1+x^2)} dx \\
 &= \int_1^{\sqrt{3}} \left(\frac{1}{x} - \frac{x}{1+x^2} \right) dx \\
 &= \left[\ln x - \frac{1}{2} \ln(1+x^2) \right]_1^{\sqrt{3}} \\
 &= \left[\ln \frac{x}{\sqrt{1+x^2}} \right]_1^{\sqrt{3}} \\
 &= \ln \left(\frac{\sqrt{3}}{\sqrt{1+3}} \right) - \ln \left(\frac{1}{\sqrt{1+1^2}} \right) \\
 &= \ln \frac{\sqrt{3}}{2} - \ln \frac{1}{\sqrt{2}} \\
 &= \ln \sqrt{\frac{3}{2}}
 \end{aligned}$$

(c) i. (2 marks)

- ✓ [1] for correctly finding $[uv]_0^1$.
- ✓ [1] for final result shown.

$$\begin{aligned}
 I_n &= \int_0^1 x^n \tan^{-1} x \, dx \\
 & \left| \begin{array}{ll} u = \tan^{-1} x & v = \frac{x^{n+1}}{n+1} \\ du = \frac{1}{1+x^2} & dv = x^n \end{array} \right. \\
 I_n &= [uv]_0^1 - \int_0^1 v \, du \\
 &= \left[\frac{x^{n+1}}{n+1} \tan^{-1} x \right]_0^1 - \int_0^1 \frac{1}{1+x^2} \left(\frac{x^{n+1}}{n+1} \right) dx \\
 &= \frac{1^{n+1}}{n+1} \tan^{-1} 1 - \frac{1}{n+1} \int_0^1 \frac{x^{n+1}}{1+x^2} dx \\
 &= \left(\frac{1}{n+1} \right) \frac{\pi}{4} - \frac{1}{n+1} \int_0^1 \frac{x^{n+1}}{1+x^2} dx
 \end{aligned}$$

Multiplying both sides by $n+1$:

$$(n+1)I_n = \frac{\pi}{4} - \int_0^1 \frac{x^{n+1}}{1+x^2} dx \quad (15.1)$$

ii. (1 mark)

When $n = 0$,

$$\begin{aligned}
 (0+1)I_0 &= \frac{\pi}{4} - \int_0^1 \frac{x^1}{1+x^2} dx \\
 &= \frac{\pi}{4} - \frac{1}{2} \left[\ln(1+x^2) \right]_0^1 \\
 &= \frac{\pi}{4} - \frac{1}{2} \ln 2
 \end{aligned}$$

iii. (2 marks)

- ✓ [1] for arriving at an expression for I_{n+2}
- ✓ [1] for final result.

Increment n to $n+2$:

$$\begin{aligned}
 (n+2+1)I_{n+2} &= \frac{\pi}{4} - \int_0^1 \frac{x^{n+2+1}}{1+x^2} dx \\
 (n+3)I_{n+2} &= \frac{\pi}{4} - \int_0^1 \frac{x^{n+3}}{1+x^2} dx \quad (15.2)
 \end{aligned}$$

Adding (15.1) and (15.2),

$$\begin{aligned}
 & (n+3)I_{n+2} + (n+1)I_n \\
 &= \frac{\pi}{4} - \int_0^1 \frac{x^{n+3}}{1+x^2} dx \\
 & \quad + \frac{\pi}{4} - \int_0^1 \frac{x^{n+1}}{1+x^2} dx \\
 &= \frac{\pi}{2} - \int_0^1 x^{n+1} \left(\frac{x^2+1}{1+x^2} \right) dx \\
 &= \frac{\pi}{2} - \int_0^1 x^{n+1} dx \\
 &= \frac{\pi}{2} - \frac{1}{n+2} \left[x^{n+2} \right]_0^1 \\
 &= \frac{\pi}{2} - \frac{1}{n+2}
 \end{aligned}$$

iv. (2 marks)

- ✓ [1] for calculating $3I_2$.
- ✓ [1] for final result.

When $n = 0$,

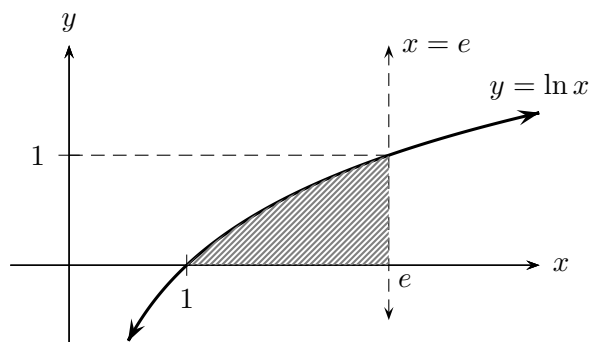
$$3I_2 + I_0 = \frac{\pi}{2} - \frac{1}{2}$$

As $I_0 = \frac{\pi}{4} - \frac{1}{2} \ln 2$,

$$\begin{aligned}
 3I_2 + \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 \right) &= \frac{\pi}{2} - \frac{1}{2} \\
 3I_2 &= \frac{\pi}{4} - \frac{1}{2} + \frac{1}{2} \ln 2 \\
 &= \frac{\pi}{4} + \frac{1}{2} (\ln 2 - 1)
 \end{aligned}$$

When $n = 2$,

$$\begin{aligned}
 5I_4 + 3I_2 &= \frac{\pi}{2} - \frac{1}{4} \\
 5I_4 &= \frac{\pi}{2} - \frac{1}{4} - \frac{\pi}{4} - \frac{1}{2}(\ln 2 - 1) \\
 &= \frac{\pi}{4} - \frac{1}{2} \left(\ln 2 - 1 + \frac{1}{2} \right) \\
 &= \frac{\pi}{4} - \frac{1}{2} \left(\ln 2 - \frac{1}{2} \right) \\
 \therefore I_4 &= \frac{\pi}{20} - \frac{1}{10} \left(\ln 2 - \frac{1}{2} \right)
 \end{aligned}$$



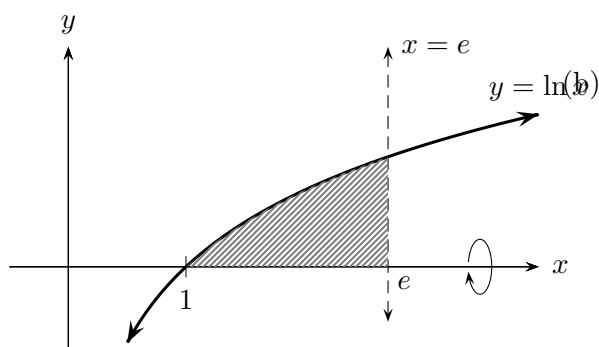
Use the area about the y axis:

Question 16 (Lam)

(a) (3 marks)

- ✓ [1] for correct application of integration by parts.
- ✓ [1] for correct finding $\int_1^e \ln x \, dx$ (graphically or by parts).
- ✓ [1] for final answer.

$$\begin{aligned}
 \int_1^e \ln x \, dx &= (e \times 1) - \int_{y=0}^{y=1} e^y \, dy \\
 &= e - [e - 1] \\
 &= 1 \\
 \therefore V &= \pi(e - 2)
 \end{aligned}$$



i. (1 mark)

Let $\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$,

$$\begin{aligned}
 \left| \underline{r} - (3\underline{i} + \underline{j} + 4\underline{k}) \right| &= \sqrt{35} \\
 \left| (x - 3)\underline{i} + (y - 1)\underline{j} + (z - 4)\underline{k} \right| &= \sqrt{35} \\
 \therefore (x - 3)^2 + (y - 1)^2 + (z - 4)^2 &= 35
 \end{aligned}$$

$$\begin{aligned}
 V &= \pi \int_1^e y^2 \, dx \\
 &= \pi \int_1^e (\ln x)^2 \, dx
 \end{aligned}$$

Finding the integral by parts/inserting 'phantom' term

$$\begin{aligned}
 &\left| \begin{array}{ll} u = (\ln x)^2 & dv = 1 \\ du = 2 \ln x \times \frac{1}{x} & v = x \\ & = \frac{2}{x} \ln x \end{array} \right. \\
 V &= \pi \left(\left[x (\ln x)^2 \right]_1^e - \int_1^e \frac{2}{x} \ln x \times x \, dx \right) \\
 &= \pi \left(e - 2 \int_1^e \ln x \, dx \right)
 \end{aligned}$$

ii. (3 marks)

- ✓ [1] for correct expression for $\left| \underline{r} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right|$ in terms of λ
- ✓ [1] for quadratic in terms of λ
- ✓ [1] for final justification.

$$\underline{r} = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

If the line is a tangent, then only one unique value of λ exists for the expression

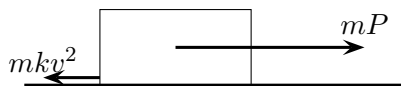
$$\begin{aligned} & \left| \underline{r} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right| = \sqrt{35} \\ & \left| \underline{r} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right| \\ &= \left| \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right| \\ &= \left| \begin{pmatrix} -1 + \lambda \\ -1 + 2\lambda \\ -1 - \lambda \end{pmatrix} \right| \\ &= \sqrt{(\lambda - 1)^2 + (2\lambda - 1)^2 + (-1 - \lambda)^2} \\ &= \sqrt{(\lambda^2 - 2\lambda + 1) + (4\lambda^2 - 4\lambda + 1) + (\lambda^2 + 2\lambda + 1)} \\ &= \sqrt{6\lambda^2 - 4\lambda + 3} = \sqrt{35} \\ &\quad \therefore 6\lambda^2 - 4\lambda + 3 = 35 \\ &\quad 6\lambda^2 - 4\lambda - 32 = 0 \\ &\quad 3\lambda^2 - 2\lambda - 16 = 0 \end{aligned}$$

Check the quadratic discriminant on λ :

$$\Delta = (-2)^2 - 4(-16)(3) > 0$$

As the discriminant is positive, there are two unique values of λ for which the line $\underline{r}$ will 'touch' the sphere. Hence it is not a tangent as it will intersect the sphere twice.

(c) i. (1 mark)



$$\begin{aligned} \sum \underline{F} &= \underline{\ddot{x}} = \underline{P} - \underline{kv^2} \\ \ddot{x} &= P - kv^2 \\ v \frac{dv}{dx} &= P - kv^2 \\ \frac{dv}{dx} &= \frac{P}{v} - kv \end{aligned}$$

ii. (3 marks)

- ✓ [1] for finding the primitive of both sides.
- ✓ [1] for showing the v^2 and x relationship.
- ✓ [1] for showing $v_M = \sqrt{\frac{P}{k}}$.

$$\frac{dv}{dx} = \frac{P - kv^2}{v}$$

Separating variables, and integrating:

$$\begin{aligned} \int \frac{v}{P - kv^2} dv &= \int dx \\ -\frac{1}{2k} \int \frac{-2kv}{P - kv^2} dv &= \int dx \\ -\frac{1}{2k} \ln(P - kv^2) &= x + C_1 \end{aligned}$$

When $x = 0$, $t = 0$ and $v = 0$:

$$-\frac{1}{2k} \ln P = 0 + C_1$$

$$C_1 = -\frac{1}{2k} \ln P$$

$$\therefore -\frac{1}{2k} \ln(P - kv^2) = x - \frac{1}{2k} \ln P$$

$$\ln(P - kv^2) = -2kx + \ln P$$

$$P - kv^2 = e^{-2kx + \ln P} = Pe^{-2kx}$$

$$kv^2 = P - Pe^{-2kx}$$

$$v^2 = \frac{P}{k} (1 - e^{-2kx})$$

As time passes, the block continues to move to the right, $x \rightarrow \infty$ and $e^{-2kx} \rightarrow 0$. Hence

$$v^2 \rightarrow \frac{P}{k}$$

$$\therefore v_M \rightarrow \sqrt{\frac{P}{k}}$$

iii. (1 mark)

At $x = x_1$, need $v^2 = \frac{1}{9} \frac{P}{k}$ (one third of the maximum speed).

$$\frac{1}{9} \frac{P}{k} = \frac{P}{k} (1 - e^{-2kx_1})$$

$$\frac{1}{9} = 1 - e^{-2kx}$$

$$e^{-2kx} = \frac{8}{9}$$

$$x_1 = -\frac{1}{2k} \ln \frac{8}{9} = \frac{1}{2k} \ln \frac{9}{8}$$

At $x = x_2$, need $v^2 = \frac{1}{4} \frac{P}{k}$ (one half of the maximum speed).

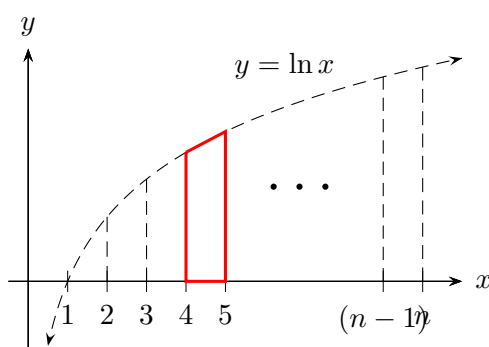
$$\begin{aligned}\frac{1}{4} \frac{P}{k} &= \frac{P}{k} (1 - e^{-2kx_2}) \\ \frac{1}{4} &= 1 - e^{-2kx_2} \\ e^{-2kx_2} &= \frac{3}{4} \\ -2kx_2 &= \ln \frac{3}{4} \\ x_2 &= \frac{1}{2k} \ln \frac{4}{3}\end{aligned}$$

Dividing,

$$\begin{aligned}\frac{x_1}{x_2} &= \frac{\frac{1}{2k} \ln \frac{9}{8}}{\frac{1}{2k} \ln \frac{4}{3}} \\ &= 0.4098 \dots \approx 41\%\end{aligned}$$

(d) i. (2 marks)

- ✓ [1] for reaching a generalisation for the area of trapeziums.
- ✓ [1] for showing the required result.
- $y = \ln x$ is concave down $\forall x$ within its domain, hence the trapeziums formed by joining the function values will be less than the actual area,



- Area of trapezium between $x = 1$ and $x = 2$:

$$A = \frac{1}{2}(1)(\ln 1 + \ln 2) = \frac{\ln 1 + \ln 2}{2}$$

- Area of trapezium between $x = 2$ and $x = 3$:

$$A = \frac{1}{2}(1)(\ln 2 + \ln 3) = \frac{\ln 2 + \ln 3}{2}$$

- Area of trapezium between $x =$

$(n-1)$ and $x = n$:

$$\begin{aligned}A &= \frac{1}{2}(1)(\ln(n-1) + \ln n) \\ &= \frac{\ln(n-1) + \ln n}{2}\end{aligned}$$

Adding all of these together,

$$\begin{aligned}\frac{\ln 1 + \ln 2}{2} + \frac{\ln 2 + \ln 3}{2} + \dots + \frac{\ln(n-1) + \ln n}{2} \\ < \int_1^n \ln x \, dx\end{aligned}$$

ii. (3 marks)

- ✓ [1] for compacting the expression to $\ln(n!) - \frac{1}{2} \ln n$.
- ✓ [1] for evaluating $\int_1^n \ln x \, dx$ (can use Q16(a) if necessary)
- ✓ [1] for final result.

Using the result above,

$$\begin{aligned}\frac{\ln 1 + \ln 2}{2} + \frac{\ln 2 + \ln 3}{2} + \dots + \frac{\ln(n-1) + \ln n}{2} \\ = \ln 2 + \ln 3 + \ln 4 + \dots + \ln(n-1) + \frac{1}{2} \ln n \\ = \ln 2 + \ln 3 + \ln 4 + \dots + \ln(n-1) \\ \quad + \ln n - \frac{1}{2} \ln n \\ = \ln(2 \times 3 \times \dots \times (n-1) \times n) \\ \quad - \frac{1}{2} \ln n \\ = \ln(n!) - \frac{1}{2} \ln n\end{aligned}$$

Finding the actual area,

$$\begin{aligned}\int_1^n \ln x \, dx \\ = \left[x \ln x - x \right]_1^n \quad (\text{By parts or otherwise}) \\ = (n \ln n - n) - (-1) \\ = n \ln n - n + 1\end{aligned}$$

Hence,

$$\begin{aligned}\ln(n!) - \frac{1}{2} \ln n &< n \ln n - n + 1 \\ \ln(n!) &< n \ln n + \frac{1}{2} \ln n - n + 1 \\ &= \left(n + \frac{1}{2}\right) \ln n - n + 1 \\ &= \ln \left(n^{n+\frac{1}{2}}\right) - n + 1 \\ &= \ln \left(n^{n+\frac{1}{2}}\right) - n \ln e + \ln e \\ &= \ln \left(n^{n+\frac{1}{2}}\right) - \ln e^n + \ln e \\ &= \ln \left(\frac{e \times n^{n+\frac{1}{2}}}{e^n}\right) \\ \therefore n! &< \frac{en^{n+\frac{1}{2}}}{e^n}\end{aligned}$$